First: Full adder from 2 half adders

Half adder:

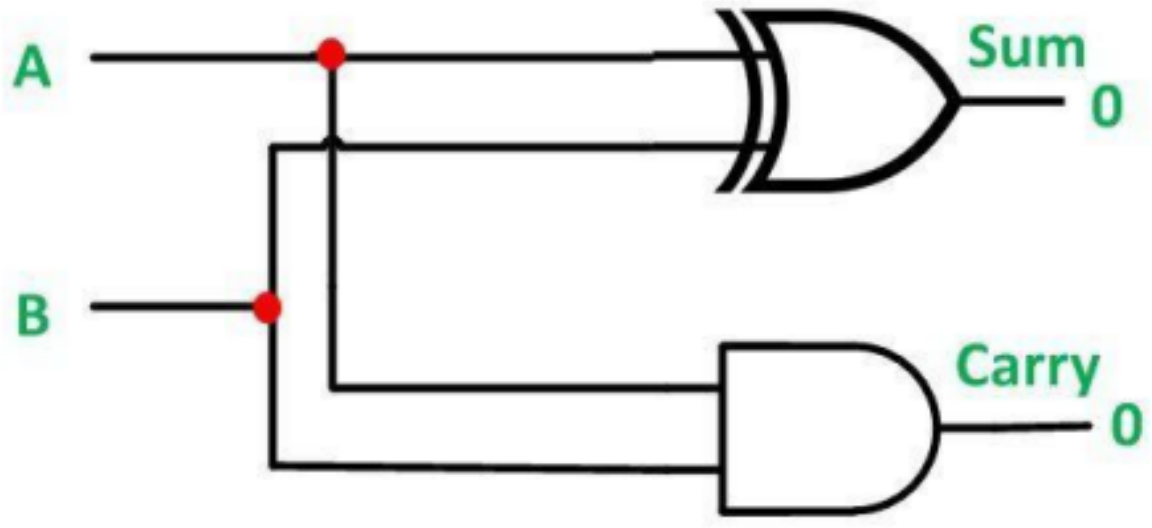
A half adder is a fundamental digital circuit used to add two single-bit binary numbers. It has two inputs (A and B) and two outputs: the sum (S) and the carry (C). The half adder adds the two input bits and produces the sum and carry as outputs.

Operation of a Half Adder:

- Sum (S): The sum output is the XOR (exclusive OR) of the two inputs. The sum is 1 if one of the inputs is 1, but not both.
- Carry (C): The carry output is the AND of the two inputs. It is 1 only if both inputs are 1, which means there is a carry-over to the next bit.

Truth Table for Half Adder:

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



Full adder:

A full adder is a more advanced digital circuit than the half adder. It adds three binary numbers: two input bits (A and B) and a carry-in bit (Cin), which comes from the previous stage in a multi-bit addition process. The full adder has two outputs: the sum (S) and the carry-out (Cout).

Operation of a Full Adder:

- Sum (S): The sum output is the XOR of all three inputs (A, B, and Cin). The sum will be 1 if an odd number of the inputs are 1.
- Carry-out (Cout): The carry-out is the OR of the AND results from the combinations of the inputs, meaning it's 1 if two or more inputs are 1.

Truth Table for Full Adder:

A	**B**	**Cin**	**Sum**	**Cout**
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Full adder from half adder:
Full adders can be implemented by using 2 half adders and an or gate.

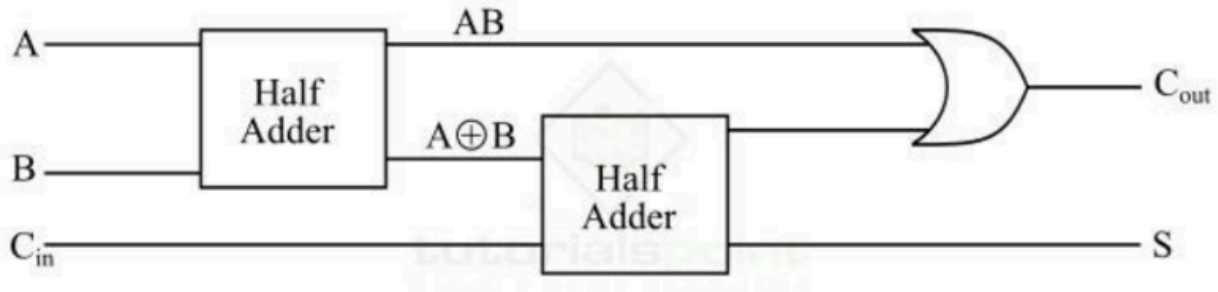
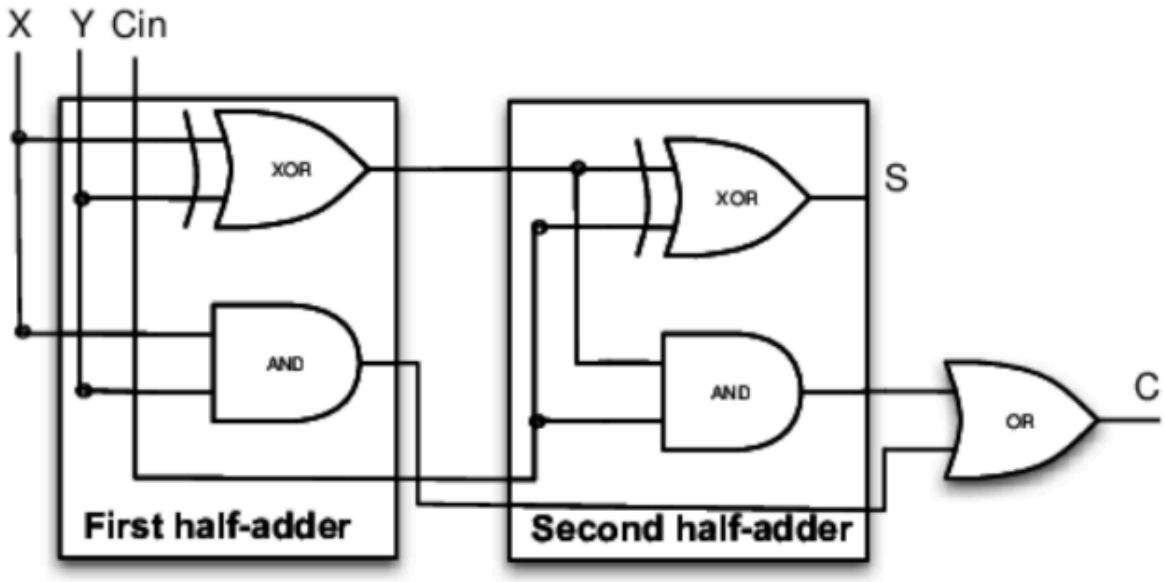


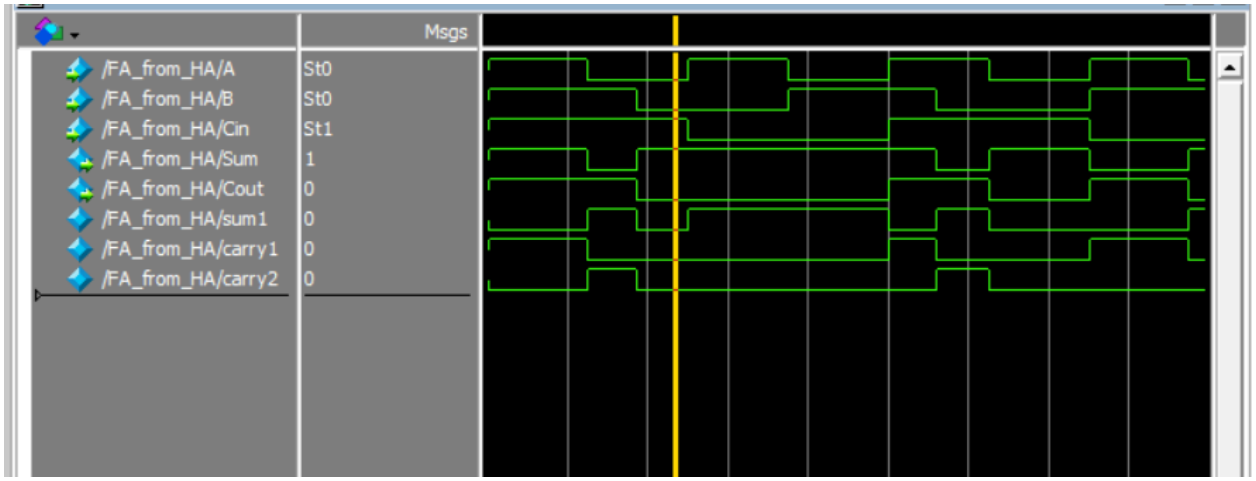
Figure 4 - Block Diagram of Full Adder using Half Adder



Code:

```
1  module HA (
2      input  logic A,
3      input  logic B,
4      output logic Sum,
5      output logic Cout
6  );
7
8      assign Sum  = A ^ B;
9      assign Cout = A & B;
10
11 endmodule
12
13 module FA_from_HA (
14     input  logic A,
15     input  logic B,
16     input  logic Cin,
17     output logic Sum,
18     output logic Cout
19 );
20     logic sum1, carry1, carry2;
21     HA ha0 (.A(A), .B(B), .Sum(sum1), .Cout(carry1));
22     HA ha1 (.A(sum1), .B(Cin), .Sum(Sum), .Cout(carry2));
23     assign Cout = carry1 | carry2;
24
25 endmodule
```

Wave form:



Full adder from two half-adders Verilog code & RTL view: